Supplemental Material

1 Analytical Results

1.1 Schwinger-Dyson Equations

The goal of this section is to derive the Schwinger-Dyson equations (10) in the main text. We begin with a simple two-state birth-death process, where the state s can take the values 0 or 1. The birth rate, γ^+, corresponds to the transition $0 \to 1$, while the death rate, γ^-, corresponds to the transition $1 \to 0$. The associated master equations are given by

$$\frac{\partial P(0,t)}{\partial t} = -\gamma^+ P(0,t) + \gamma^- P(1,t) \tag{S1}$$

$$\frac{\partial P(1,t)}{\partial t} = \gamma^+ P(0,t) - \gamma^- P(1,t). \tag{S2}$$

Given an observable $O(s)$, we have

$$\begin{aligned} \frac{d\langle O(s)\rangle}{dt} &= \sum_{s=0,1} \frac{\partial P(s,t)}{\partial t} O(s) \\ &= (O(1)-O(0))\gamma^+ P(0,t) - (O(1)-O(0))\gamma^- P(1,t). \\ &= \Big\langle \Big((1-s)\gamma^+ - s\gamma^-\Big)(O(1)-O(0)) \Big\rangle \end{aligned} \tag{S3}$$

Similarly, consider a directed graph with N vertices, described by the Boolean adjacency matrix A. We analyze the general network edge addition and removal process, characterized by the birth rate $\gamma_e^+(A)$ and death rate $\gamma_e^-(A)$ for each edge e. The evolution of the expectation of a physical observable, $\langle O(A)\rangle := \int O(A)P(A,t)dA$, satisfies

$$\frac{d\langle O(A)\rangle}{dt} = \sum_{e\in E} \Big\langle \Big((1-A_e)\gamma_e^+(A) - A_e\gamma_e^-(A)\Big)(O(A|A_e=1) - O(A|A_e=0)) \Big\rangle. \tag{S4}$$

For a connected subgraph G with $|V_G| = n$ and $|E_G| = m$, we are interested in the subgraph density $\rho(G) := \frac{1}{N}\sum_{v_1,\ldots,v_n\in V}\prod_{(i,j)\in E_G} A_{v_i,v_j}$, which is related to the enumeration of the subgraph, $N(G) = \frac{N}{n!}\rho(G)$. We assume the sparsity condition, i.e., $\rho(G)$ is finite for all subgraphs G with size $|V_G| = O(1)$.

The subgraph density can be interpreted as an analog of an equal-time n-point correlation function, with the adjacency matrix $A(t)$ acting as a field variable, much like fields in quantum field theory. Specifically, here $O(A) = \prod_{(i,j)\in E_G} A_{v_i,v_j}$ represents a product of field variables A at different positions. Moreover, for a given edge e, we find that:

$$\begin{cases} O(A|A_e=0)=0,\ O(A|A_e=1)=\prod_{(i',j')\in E_G/(i,j)} A_{v_{i'},v_{j'}}, & \text{if } e=(v_i,v_j),\ \text{for } (i,j)\in E_G, \\ O(A|A_e=0)=O(A|A_e=1)=O(A), & \text{otherwise.} \end{cases} \tag{S5}$$

In the minimal model, $\gamma^+_{ij}(A) = \alpha/N + \beta\sum_k A_{ik}A_{kj}$ and $\gamma^-_{ij}(A) = \gamma$. Substituting this and Eq. (S5) into Eq. (S4), we obtain:

$$\begin{aligned} \frac{\partial\rho(G,t)}{\partial t} = \frac{\alpha}{N^2} \sum_{(i,j)\in E_G} \sum_{v_1,\dots,v_n\in V} \left\langle (1-A_{v_i,v_j}) \prod_{(i',j')\in E_G/(i,j)} A_{v_{i'},v_{j'}} \right\rangle + \\ \frac{\beta}{N} \sum_{v_0,v_1,\dots,v_n\in V} \left\langle (1-A_{v_i,v_j}) A_{v_i,v_0} A_{v_0,v_j} \prod_{(i',j')\in E_G/(i,j)} A_{v_{i'},v_{j'}} \right\rangle - m\gamma\rho(G,t). \end{aligned} \tag{S6}$$

Note that

$$\frac{1}{N^2} \sum_{v_1,\dots,v_n\in V} \left\langle (1-A_{v_i,v_j}) \prod_{(i',j')\in E_G/(i,j)} A_{v_{i'},v_{j'}} \right\rangle = \frac{1}{N^2} \sum_{v_1,\dots,v_n\in V} \left\langle \prod_{(i',j')\in E_G/(i,j)} A_{v_{i'},v_{j'}} \right\rangle - \frac{1}{N}\rho(G,t) \tag{S7}$$

Thus, the second term vanishes in the thermodynamic limit as long as the sparsity condition holds. The first term also vanishes except when e' is a bridge of G, i.e., removing e' causes G to become a disconnected union of two graphs $G_1(e')$ and $G_2(e')$. In this case, we have $\frac{1}{N^2}\left\langle \prod_{e'\in E_G, e'\neq e} A_{e'} \right\rangle = \rho(G_1)\rho(G_2)$, which does not vanish.

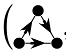
Now consider the term with β, which creates a new edge $e=(u,w)$ from vertex u to w due to a common neighbor v. We define the induced graph $I_v(e)$ for $e=(u,w)$ as $I_v(e) = \{\{u,v,w\},\{(u,v),(v,w)\}\}$. There are two cases: (i) $v_0 \notin V_G$, which leads to a term $\rho(G/e\cup I_0(e),t) - \rho(G\cup I_0(e),t)$; and (ii) $v_0 = v\in V_G$, which leads to $\sum_{v\in V_G}\rho(G/e\cup I_v(e),t) - \rho(G\cup I_v(e),t)$. Taken together, we obtain the Schwinger-Dyson equations

$$\begin{aligned} \frac{\partial\rho(G,t)}{\partial t} &= \alpha\sum_{b\in B_G}\rho(G_1(b),t)\rho(G_2(b),t) - m\gamma\rho(G,t) \\ &+\beta\sum_{e\in E_G}\left(\rho(G/e\cup I_0(e),t) - \rho(G\cup I_0(e),t) + \sum_{v\in V_G}\rho(G/e\cup I_v(e),t) - \rho(G\cup I_v(e),t)\right), \end{aligned} \tag{S8}$$

where B_G is the set of bridges. The first term on the RHS represents the formation of bridges between two disconnected subgraphs $G_1(b)$ and $G_2(b)$. The second term represents the contribution from triadic closure, accounting for whether the common neighbors lie outside (v_0) or inside (v) the vertex set V_G.

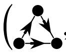
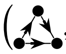
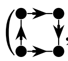
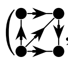
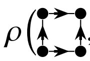
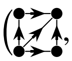
Let $G = \bullet\!\rightarrow\!\bullet$, we have

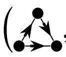
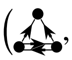
$$\frac{\partial\rho(\bullet\!\rightarrow\!\bullet,t)}{\partial t} = \alpha\rho(\bullet,t)^2 - \gamma\rho(\bullet\!\rightarrow\!\bullet,t) + \beta\Big(\rho(\bullet\!\rightarrow\!\bullet\!\rightarrow\!\bullet,t) - \rho\big(\triangle,t\big)\Big), \tag{S9}$$

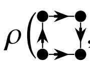
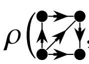
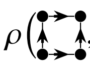
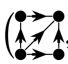
where $\rho(\bullet, t) = 1$, $\rho(\bullet\!\to\!\bullet, t) = z$, $\rho(\bullet\!\to\!\bullet\!\to\!\bullet, t) = \left\langle z^{in} z^{out} \right\rangle = z^2$, $\rho\Big($$, t\Big) = \Delta$. We then find Eq. (2) in the main text

$$\frac{\partial z}{\partial t} = \alpha - \gamma z + \beta\left(z^2 - \Delta\right). \tag{S10}$$

Let $G =$ , we have

$\dfrac{\partial \rho\Big(\right.$$\left., t\Big)}{\partial t} = -3\gamma\rho\Big($$, t\Big) + \beta\Big(2\rho\Big($$, t\Big) - 2\rho\Big($$, t\Big) + \rho\Big($$, t\Big) - \rho\Big($$, t\Big)\Big)$

$+\beta\Big(\rho(\bullet\!\to\!\bullet\!\to\!\bullet, t) - \rho\Big($$, t\Big) + \rho(\bullet\!\rightleftarrows\!\bullet\!\to\!\bullet, t) - \rho\Big($$, t\Big) + \rho(\bullet\!\to\!\bullet\!\rightleftarrows\!\bullet, t) - \rho\Big($$, t\Big)\Big).$ (S11)

It is easy to see that $\rho\Big($$, t\Big)$, $\rho\Big($$, t\Big)$, $\rho\Big($$, t\Big)$, $\rho\Big($$, t\Big)$, $\rho\Big($$, t\Big)$, and $\rho\Big($$, t\Big)$ are at least of order $O(\beta^2)$. Furthermore, $\rho(\bullet\!\rightleftarrows\!\bullet\!\to\!\bullet, t)$ and $\rho(\bullet\!\to\!\bullet\!\rightleftarrows\!\bullet, t)$ also are of order $O(\beta^2)$. To see this, we note

$$\frac{\partial \rho(\bullet\!\rightleftarrows\!\bullet\!\to\!\bullet, t)}{\partial t} = \alpha\rho(\bullet, t)\rho(\bullet\!\rightleftarrows\!\bullet, t) - 3\gamma\rho(\bullet\!\rightleftarrows\!\bullet\!\to\!\bullet, t) + O(\beta^2), \tag{S12}$$

which implies $\rho(\bullet\!\rightleftarrows\!\bullet\!\to\!\bullet, t)$ is of the same order as $\rho(\bullet\!\rightleftarrows\!\bullet, t)$. The same result can be found similarly for $\rho(\bullet\!\to\!\bullet\!\rightleftarrows\!\bullet, t)$. Moreover, we have

$\dfrac{\partial \rho(\bullet\!\rightleftarrows\!\bullet, t)}{\partial t} = -2\gamma\rho(\bullet\!\rightleftarrows\!\bullet, t) + \beta\Big(\rho\Big($$, t\Big) - \rho\Big($$, t\Big)\Big),$ (S13)

which shows $\rho(\bullet\!\rightleftarrows\!\bullet, t) \sim O(\beta^2)$. Taken together, we recover Eq. (3) in the main text, as

$$\frac{\partial \Delta(t)}{\partial t} = -3\gamma\Delta + \beta(z^2 - \Delta) + O(\beta^2). \tag{S14}$$

At the tree level, where we ignore the contributions of Δ, incorporating Eq. (S10) with (S14) leads to

$$\beta z(t) = \begin{cases} \dfrac{2\omega}{1 - \exp\left[2\sqrt{\beta}\omega t + \ln\left(\frac{\frac{1}{2}+\omega}{\frac{1}{2}-\omega}\right)\right]} + \frac{1}{2} - \omega & (\beta < \frac{\gamma^2}{4\alpha}), \\ \frac{1}{2} - \frac{\beta}{2\beta + t} & (\beta = \frac{\gamma^2}{4\alpha}), \\ \frac{\gamma}{2} + \omega\cot(\omega(t_c - t)) & (\beta > \frac{\gamma^2}{4\alpha}), \end{cases} \tag{S15}$$

for the initial condition $z(0) = 0$. Here, $\omega = \sqrt{|\alpha\beta - \frac{\gamma^2}{4}|}$ and $t_c = \omega^{-1}(\mathrm{acot}(2\omega) + \pi/2)$. A similar result can be obtained for arbitrary initial conditions $z(t = 0)$. It is straightforward to verify the asymptotic behavior in Eq. (S10) explicitly.

1.2 Degree Distribution

In this section, we present the analytical solution for the time-dependent degree distribution at the tree level. We start with the master equation:

$$\frac{\partial P(k,t)}{\partial t} = (\alpha + \beta z(t)(k-1))P(k-1,t) - (\alpha + (\beta z(t) + \gamma)k)P(k,t) + \gamma(k+1)P(k+1,t), \tag{S16}$$

which describes the evolution of the degree distribution. Here, we omit higher-order corrections such as terms involving Δ. An exact form of $P(k,t)$ can be obtained using the Schwinger-Dyson equations (S8).

Equation (S16) can be solved analytically by introducing the generating function $G(x,t) := \sum_{k=0}^{\infty} x^k P(k,t)$. Substituting the $G(x,t)$ into the Eq. (S16) yields the partial differential equation

$$\frac{\partial G(x,t)}{\partial t} = (x-1)\left(\alpha G(x,t) - (\gamma - \beta z(t)x)\frac{\partial G(x,t)}{\partial x})\right). \tag{S17}$$

To solve Eq.(S17), we consider

$$\frac{\partial G(x,t)}{\partial t} + (x-1)((x-1)f(t)+g(t))\frac{\partial G(x,t)}{\partial x} = \alpha(x-1)G(x,t), \tag{S18}$$

where $f(t) = -\beta z(t)$ and $g(t) = \gamma - \beta z(t)$. It is convenient to introduce

$$\tilde{G}(X(x,t),T(t)) := G(x,t), \tag{S19}$$

where $X(x,t) := \frac{E(t)}{x-1}$, with $E(t) := \exp\left(\int_0^t g(t')dt'\right)$ and $T(t) := \int_0^t f(t')E(t')dt'$. Substituting into Eq. (S17), we have

$$X\tilde{f}(T)(\partial_T \tilde{G} - \partial_X \tilde{G}) = \alpha\tilde{G}, \tag{S20}$$

where $\tilde{f}(T(t)) := f(t)$. The solution is given by

$$\tilde{G}(X,T) = \exp\left(\alpha\int_0^T \frac{1}{\tilde{f}(T')(X+T-T')}dT'\right)\Phi(X+T), \tag{S21}$$

where $\Phi(\cdot)$ is an arbitrary function determined by the initial condition. Consequently, we obtain

$$G(x,t) = \exp\left(\alpha(x-1)\int_0^t \frac{E(t')}{E(t)+(T(t)-T(t'))(x-1)}dt'\right)G_0\left(\frac{x-1}{T(t)(x-1)+E(t)}+1\right), \tag{S22}$$

where $E(t) := \exp\left(\gamma t - \beta\int_0^t z(t')dt'\right)$ and $T(t) := \int_0^t(-\beta z(t'))E(t')dt = E(t) - \gamma\int_0^t E(t')dt - 1$, and $G_0(x)$ is the probability generating function at $t = 0$. We assume that the initial graph is empty for simplicity, thus $G_0(x) = 1$. However, similar results can be obtained for general initial conditions.

To obtain the degree distribution $P(k,t)$ from $G(x,t)$, we note that the generating function at $x = e^{i\phi}$, $G(e^{i\phi},t) = \sum_k e^{ik\phi}P(k,t)$, corresponds to the Fourier series transform of $P(k,t)$. Specifically, we have

$$G(e^{i\phi},t) = \exp\left(\alpha(e^{i\phi}-1)\int_0^t \frac{E(t')}{E(t)+(T(t)-T(t'))(x-1)}dt'\right). \tag{S23}$$

Therefore, the degree distribution $P(k,t)$ can be obtained by the inverse Fourier transform

$$P(k,t) = \int_{-\infty}^{\infty} \mathrm{d}\phi e^{-i2\pi k\phi}G(e^{i\phi},t), \tag{S24}$$

as shown in Fig. 4a of the main text for various t values. To further compute the fraction of isolated vertices p_0 at criticality, we use $p_0 = P(0,t_c) = G(0,t_c)$, obtaining

$$p_0 = \exp\left(-\alpha\int_0^{t_c} \frac{E(t')}{E(t)-(T(t_c)-T(t'))}dt'\right). \tag{S25}$$

1.3 Stationary States and Exponential Random Graph Model (ERGM)

To clarify the connection between our dynamical birth–death process and equilibrium exponential random graph models (ERGMs), we analyze in detail the case of an ERGM with edge and triangle contributions. This allows us to establish explicitly under what conditions detailed balance is satisfied, and how our linear-rate dynamics provides an approximate but genuinely non-equilibrium realization.

Consider an ERGM with both edge and triangle contributions,

$$\pi(G) \propto \exp(\theta_1 \, \#\text{edges}(G) + \theta_2 \, \#\text{triangles}(G)). \tag{S26}$$

A $0 \to 1$ toggle at (i, j) changes the exponent by

$$\Delta E = \theta_1 + \theta_2 \, (A^2)_{ij}. \tag{S27}$$

By definition, the detailed balance satisfies the exponential form

$$\frac{W(0 \to 1)}{W(1 \to 0)} = e^{\Delta E}, \tag{S28}$$

so detailed balance is satisfied exactly if one chooses an *exponential* birth rate

$$\frac{\alpha_{ij}}{\gamma} = \exp(\theta_1 + \theta_2 \, (A^2)_{ij}). \tag{S29}$$

In this case, the Kolmogorov cyclic criteria is satisfied, that is,

$$\prod_{(i,j)\in A} \frac{W(0 \to 1)}{W(1 \to 0)} = e^{\sum_{(i,j)} \Delta E_{ij}} = 1. \tag{S30}$$

Expanding Eq.(S29) via a Taylor expansion yields

$$\exp(\theta_1 + \theta_2 \, (A^2)_{ij}) = \exp^{\theta_1}(1 + \theta_2 (A^2)_{ij} + (\theta_2 (A^2)_{ij})^2/2 + \ldots). \tag{S31}$$

An accurate approximation in the weak-triadic limit can be written as

$$W(0 \to 1) \approx \gamma \exp^{\theta_1} + \gamma \exp^{\theta_1} \theta_2 (A^2)_{ij}. \tag{S32}$$

If instead a *linear* birth rate is used, $\alpha_{ij} = \alpha + \beta (A^2)_{ij}$, then matching to first order (small $|\theta_2 (A^2)_{ij}|$) via a Taylor expansion yields

$$\alpha = \gamma e^{\theta_1}, \qquad \beta = \gamma e^{\theta_1} \theta_2, \qquad \text{so} \quad \frac{\beta}{\alpha} \approx \theta_2. \tag{S33}$$

Equations (S33) establish the approximate mapping between our linear-rate birth–death dynamics and the ERGM parameters in the weak-coupling limit. Importantly, however, only the exponential-rate form [Eq. (S28)] exactly satisfies the Kolmogorov cycle criterion and hence detailed balance. By contrast, the linear-rate version used in our model cannot in general be written in exponential form, and thus violates detailed balance except in the trivial $\beta \to 0$ case.

Therefore, while our dynamics shares structural similarities with ERGMs and recovers them in certain limits, it remains fundamentally out of equilibrium. Even in steady states, the system is in a non-equilibrium steady state with positive entropy production.

2 Numerical Results

2.1 Estimation of the critical time t_c

We estimate t_c using two complementary procedures: (a) *derivative-based*, by locating the time of steepest ascent of $z(t)$ (maximum of a smoothed $\frac{dz}{dt}$, consistent with $\frac{dz}{dt} \to \infty$ as $t \to t_c$); and (b) *fit-based*, by nonlinear least-squares fitting of pre-transition data to the theoretical form (either the asymptotic $z(t) \approx A/(t_c - t) + B$ or the full solution in Eq. (S15)). Both procedures yield consistent estimates of t_c across parameter sweeps. Figure S1 shows that the extracted t_c closely matches the analytical prediction for varied α, β, and γ.

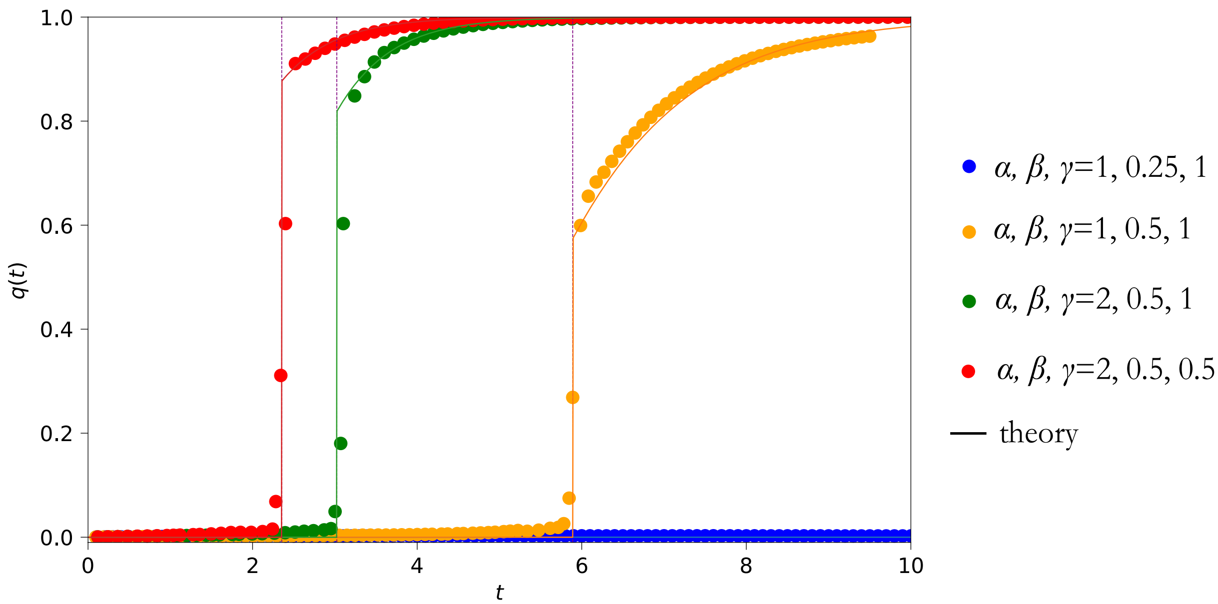


Figure S1: **Dependence of $q(t)$ and t_c on model parameters.** Increasing α or β leads to an earlier and stronger DPT, while larger γ delays the transition and weakens its sharpness. The agreement between theory and simulations is high, with $R^2 = 0.89$ (orange), 0.93 (green), and 0.94 (red).

2.2 Verification of theoretical predictions

We systematically varied (α,β,γ) and quantified the response of the order parameter and t_c. Increasing α or β reduces t_c and sharpens the pre-transition growth, whereas increasing γ delays the transition (larger t_c) and softens the growth. These trends are fully consistent with the analytical solution Eq. (S15) and the threshold $\beta_c = \gamma^2/(4\alpha)$. For the three representative explosive cases in Fig. S1, the agreement between theory and simulation is excellent ($R^2 = 0.89, 0.93, 0.94$).

2.3 Convergence in the thermodynamic limit

The apparent smoothness of simulated $q(t)$ is fully attributable to finite-size effects: in finite networks, fluctuations and averaging smear the abrupt jump predicted in the $N \to \infty$ limit. As shown in Fig. S2, increasing the system size N leads to progressively steeper transitions, and the simulation curves converge toward the theoretical discontinuity at t_c. For $N = 800$, finite-size effects are already negligible, and the simulations show excellent quantitative agreement with the theoretical prediction, demonstrating convergence toward the $N \to \infty$ limit.

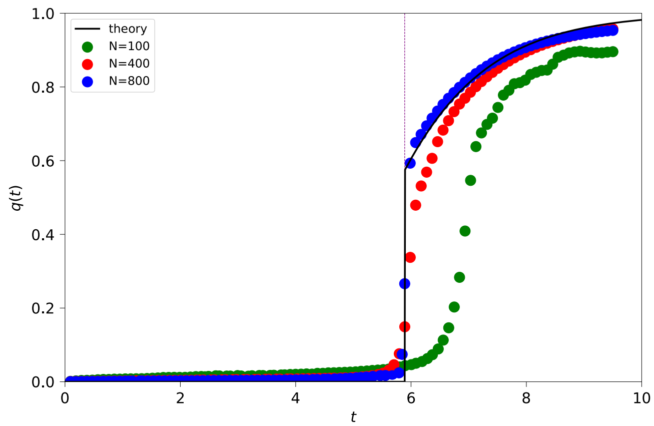


Figure S2: **Finite-size effects.** Simulations with increasing system size N under the same control parameters. As N grows, the transition sharpens and the simulated $q(t)$ approaches the theoretical discontinuity at t_c.